Final Exam

Sam Olson

Overall Evaluation - Thoughts

Short Answer Questions

1.

In lecture, we discussed an example taken from the book *Generalized Linear Models* by McCullagh and Nelder that illustrated the use of basic quasi-likelihood. In that example, use was made of Pearson residuals,

$$r_i = \frac{(y_i - \hat{\mu}_i)}{[\mathrm{var}(\hat{\mu}_i)]^{1/2}}.$$

Why are Pearson residuals better suited for use with quasi-likelihood than are deviance residuals?

Answer - Me

Answer - AI

Evaluation

2.

In any problem involving censored observations for which we will use a likelihood that contains terms for left censored observations and/or interval censored observations and/or right censored observations, what is a major assumption that affects the validity of the analysis?

Answer - Me

Answer - AI

Evaluation

3.

There is some evidence that Fisher may have considered aspects of a testing problem such as normality, constant variance, and perhaps even independence as part of the hypothesis to be tested. This would not be possible under a Neyman-Pearson framework in which these aspects of the test must be given as assumptions under which the test is valid. Why?

Answer - Me

Answer - AI

Evaluation

4.

What is the purpose of using what in class was called a comparison function $h(\theta, \hat{\theta}_n)$ in a parametric bootstrap?

Answer - Me

Answer - AI

Evaluation

5.

Why are normed profile likelihoods of no help in finding maximum likelihood estimates?

Answer - Me

Answer - AI

Evaluation

6.

Assessment of a Bayesian analysis is often approached through use of the posterior predictive distribution for new observations,

$$p(\mathbf{y}^0 \mid \mathbf{y}) = \int f(\mathbf{y}^0 \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{y})\, d\boldsymbol{\theta}.$$

(a)

For this definition to hold, $\mathbf{Y}^0$ must follow the same distribution as $\mathbf{Y}$, depending on the same value of $\boldsymbol{\theta}$. What else must be true about the relation between $\mathbf{Y}^0$ and $\mathbf{Y}$ for this definition of a posterior predictive distribution to be valid?

Answer - Me

Answer - AI

Evaluation

(b)

There are two other predictive distributions that can be useful in a Bayesian analysis. One can be used with the standard data model and prior setup consisting of a data model $f(\mathbf{y}|\boldsymbol{\theta})$ and a prior $\pi(\boldsymbol{\theta})$. The other is connected specifically with hierarchical models consisting of $f(\mathbf{y}|\boldsymbol{\theta})$, $g(\boldsymbol{\theta}|\boldsymbol{\lambda})$ and $\pi(\boldsymbol{\lambda})$. Define these predictive distributions, and indicate their primary uses.

Answer - Me

Answer - AI

Evaluation

Additional Questions

1.

A set of environmental samples (say soil samples) have been taken to determine the level of contamination of a site by polychlorinated biphenyls (PCBs). Two laboratories were used to analyze the samples, with n_1 samples being sent to Lab 1 and n_2 samples sent to Lab 2. Lab 1 has a detection limit of 0.01 μg/L, while Lab 2 has a detection limit of 0.03 μg/L. Both labs have a precision of 0.005 μg/L for values reported to two decimal places. The data recorded are laboratory (1 or 2), sample number, concentration in μg/L, and an indicator as to whether the sample was below detection limits (1) or not (0). We have decided to model concentrations with a gamma distribution, parameterized for a random variable X and parameters $\alpha > 0$ and $\beta > 0$ as,

$$f(x \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x); \quad x > 0.$$

Write the likelihood function for the entire collection of n_1 samples analyzed at Lab 1 and n_2 samples analyzed at Lab 2. Assume independence of samples, and note that our objective is to estimate the distribution of concentrations, NOT to compare the results between the two laboratories.

HINT: Define random variables and set up your indexing system first.

Answer - Me

Answer - AI

Evaluation

2.

Consider four random variables X_1, X_2, Y, and Z. You are told that X_1 and X_2 are conditionally independent given the pair (Y, Z) but that Y and Z are not independent. Determine two conditions, similar in form but one for X_1 and one for X_2, that are sufficient for $[X_1, X_2 \mid Y, Z] = [X_1 \mid Y][X_2 \mid Z]$.

Answer - Me

Answer - AI

Evaluation

3.

Consider a situation in which we have two scientific or substantive processes of interest. These could, for example, be two environmental processes in a rainforest such as tree growth and evapotranspiration, or two constructs in the social sciences such as access to preventative health care and social status in rural areas. We may have some scientific basis to believe these processes are not independent of one another. We also have two observable quantities that we believe are direct reflections of those processes, such as vegetative biomass and respiration in sample plots for the rainforest example, or number of rural medical centers and poverty rates in counties for the social science example.

(a)

First, define some notation to make a problem such as those described (two non-independent processes and two direct indicators) fit into the notational framework of the previous question. That is, define X_1, X_2, Y, and Z in terms of processes (say process 1 and process 2), and observables (say observable 1 and observable 2). You can, without additional notation, assume that writing down one observable random variable implies a collection of values from sampling units.

Answer - Me

Answer - AI

Evaluation

(b)

Using an extension of your notation from (a), identify the distributional forms that will be needed to formulate a hierarchical model. For example, if I desire to model the distribution of a random variable W conditional on the value of another random variable S with a density or mass function that has a parameter θ I would say I need to formulate a model for $[W \mid S, \theta]$.

Answer - Me

Answer - AI

Evaluation

(c)

Suppose we are willing to assume the conditions given and identified in your answer to the previous question. Now identify what part(s) of the model formulation from part (b) is(are) made more simple as a result of that assumption.

Answer - Me

Answer - AI

Evaluation

(d)

In one or possibly two (but one should really do it) sentence or sentences, state what the assumption implies about the actual processes and observable quantities involved in the problem. If you wish, you may use one of the hypothetical examples given in the preamble to this question to make things more concrete (but you certainly do not have to do that).

Answer - Me

Answer - AI

Evaluation

4.

If you have heard any news about the stock market over the past 10 weeks or so, you know that there have been some days with extreme losses, some with sizable "bounce-back" gains, and some with relatively little movement. This is called high "volatility" in the market. There are all kinds of ways one might model a high volatility situation, but suppose we suggest the following:

Let $\{Y_t : t = 1, \ldots, T\}$ denote random variables associated with the change in an index of the stock market on day t (change from day $t-1$). Formulate the following model

$$Y_t = \mu + \epsilon_t$$

$$\epsilon_t \sim N(0,\, S_t^2)$$

$$S_t = \alpha S_{t-1} + v_t$$

$$v_t \overset{\text{iid}}{\sim} N(0,\, \sigma^2)$$

(a)

Determine the expected values, variances, and one-step covariances of S_t (that is, the covariance of S_t with S_{t-1} or S_{t+1}).

Hint: If you recall them, you may use the results we obtained in class for AR(1) processes without derivation.

Answer - Me

Answer - AI

Evaluation

(b)

Determine the expected values, variances, and one-step covariances of ϵ_t.

Hint: Use of the standard conditioning argument for expected values and variances might be valuable here.

Answer - Me

Answer - AI

Evaluation

(c)

Determine the expected values, variances, and one-step covariances of Y_t.

Answer - Me

Answer - AI

Evaluation

(d)

What does this model then imply about a sequence of changes in the index of stock market prices over the given time period?

Note: Just for information (but not really important for this question), an example series of values $Y_1, \ldots, Y_T$ *with* $T = 100$, $\mu = 0$, $\sigma^2 = 1.5$ *and* $\alpha = 0.90$ *is shown in Figure 1.*

Answer - Me

Answer - AI

Evaluation

5.

Consider a formal Goodness of Fit test, such as the Chi-squared test of Karl Pearson. Briefly discuss why Pearson's notion of testing goodness of fit does not fit squarely into either the Fisherian theory of significance tests, or the Neyman-Pearson theory of hypothesis tests.

Answer - Me

Answer - AI

Evaluation

6.

Consider two continuous, independent random variables X and Y that have distributions with densities $p(x|\theta)$ and $p(y|\lambda)$, respectively, with each having support on the entire real line. Suppose we would like to compute a Monte Carlo approximation to $\Pr(Y > X)$ for given values of θ and λ.

There are two Monte Carlo approximations to $\Pr(Y > X)$ that immediately suggest themselves. Both procedures are based on a sample of values $\{(x_m, y_m) : m = 1, \ldots, M\}$ where the x_m are simulated from $p(x|\theta)$ and the y_m are independently simulated from $p(y|\lambda)$. Approximation A is,

$$P_M^{(A)}(Y > X) = \frac{1}{M} \sum_{m=1}^{M} I(y_m > x_m),$$

where $I(\cdot)$ is the usual indicator function. Approximation B is

$$\overset{(B)}{\underset{M}{\Pr}}(Y > X) = \frac{1}{M} \sum_{m=1}^{M} \frac{1}{M} \sum_{j=1}^{M} I(y_j > x_m).$$

(a)

Identify the explicit integral forms, both equal to $\Pr(Y > X)$, that $P_M^{(A)}$ and $P_M^{(B)}$ are approximating. That is, there are any number of integral expressions that are equal to $\Pr(Y > X)$. You don't get credit for some form just because it is equal to $\Pr(Y > X)$. What we want are those forms that translate directly into the Monte Carlo approximations A and B.

Answer - Me

Answer - AI

Evaluation

(b)

Identify, in terms of expectation operators, again both equal to $\Pr(Y > X)$, the expected values that $P_M^{(A)}$ and $P_M^{(B)}$ are approximating.

Hint: Just because X and Y are independent, that does not preclude us from thinking about the conditional distribution of one given the other. It just means that $p(x|y) = p(x)$, and similarly for Y.

Note About the Question: While these approximations are asymptotically equivalent, there can be differences in Monte Carlo variance for finite Monte Carlo sample sizes, M. In particular, approximation B seems to have a smaller Monte Carlo error than approximation A. Even more interesting, the variances have a nearly constant ratio $V^{(B)}/V^{(A)}$ of about 0.65 to 0.70 as $M \to \infty$ (giving relative efficiency), although both variances are converging to 0, of course.

Answer - Me

Answer - AI

Evaluation